

Guía para la guapa

Requisitos tener un wsl Ubuntu 2.4 o 2.2 o Linux native or a Dual boot

2.2 for Humble (Humble todavía es una distribución activa)

2.4 for Jazzy (Probably, next year this distribution will be the most efficient; also, the teacher said this is the easiest distribution for beginners)

Preguntas de la clase:

¿Qué es el ARK OPEN CHALLENGEr ROBOTICS?

El track del Torneo mexicano de robótica, ¿es el de carros autónomos?

Ros2 usa nodos

Links de la clase:

Guide of WSL https://learn.microsoft.com/en-us/windows/wsl/install

Guide to download Humble:

https://docs.ros.org/en/humble/Installation/Ubuntu-Install-Debs.html

Installation guide of Jazzy **(We need to follow this)**

https://docs.ros.org/en/jazzy/Installation.html

For some reason, I need to use the next link to solve a problem related to ‘gz’ this is optional

https://gazebosim.org/docs/harmonic/install_ubuntu/

After you install WSL

Go to the https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html guide

Follow all the steps except ROS-Base Install (Bare Bones): Communication libraries, message packages, command line tools. No GUI tools.

Desktop Install (Recommended): ROS, RViz, demos, tutorials.

```
$ sudo apt install ros-humble-desktop
```

ROS-Base Install (Bare Bones): Communication libraries, message packages, command line tools. No GUI tools.

```
$ sudo apt install ros-humble-ros-base
```

Development tools: Compilers and other tools to build ROS packages

```
sudo apt install ros-dev-tools
```

In this case, the Development tools are necessary. Don't include the last environment line, but this line will be useful in the next command

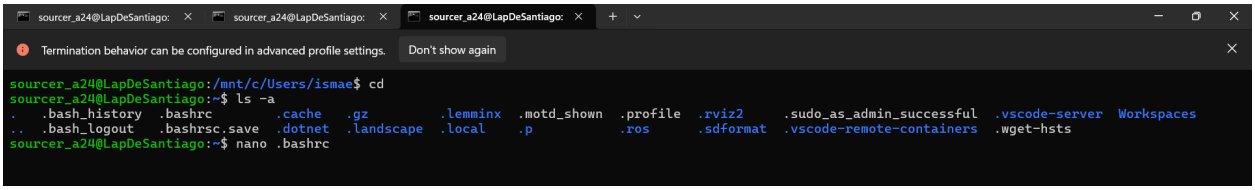
Environment setup

Sourcing the setup script

Set up your environment by sourcing the following file.

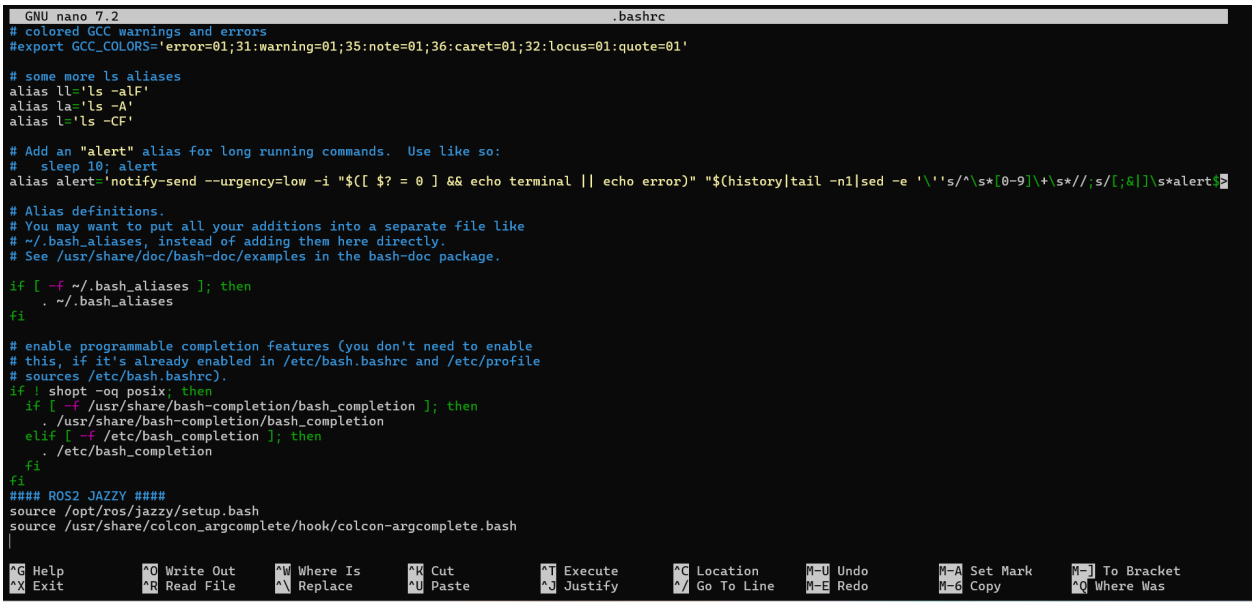
```
$ source /opt/ros/humble/setup.bash
```

In the guide are not included this information, but you must set up the environment every time you use ROS2, unless you do the next step.



This will open the nano text editor

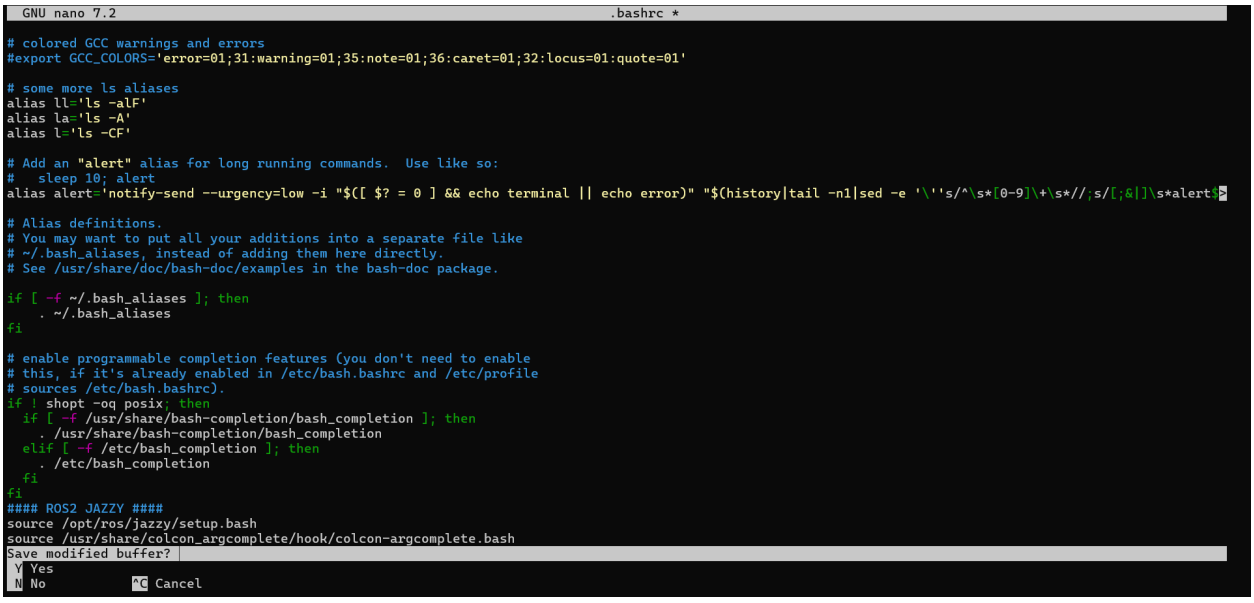
In the last line of the file, we need to put this



code:

```
####ROS2 JAZZY####
source /opt/ros/jazzy/setup.bash
source /usr/share/colcon_argcomplete/hook/colcon-argcomplete.bash
```

Then we need to save them



Press Y and next enter

To apply these effects, we need to put the command source and the name of the file

If everything is good, you shouldn't have problems displayed on the console

```
sourcer_a24@LapDeSantiago:~$ source .bashrc
sourcer_a24@LapDeSantiago:~$
```

Note: If you have a dedicated GPU, you can add other commands just before the last code:

```
# enable using nvidia gpu for graphics
export MESA_D3D12_DEFAULT_ADAPTER_NAME=NVIDIA
#### ROS2 Jazzy ####
source /opt/ros/jazzy/setup.bash
source /usr/share/colcon_argcomplete/hook/colcon-argcomplete.bash
```

In my case, I have a GPU Intel X, but the GPU is not dedicated as an NVIDIA GPU

Note: the command that we already put in is to avoid the need to set up the environment every time we work with ROS2

After, we need to clone the repository

```
sourcer_a24@LapDeSantiago:~$ mkdir -p Workspaces/movilidad_ws/src
sourcer_a24@LapDeSantiago:~$ ls
Workspaces
sourcer_a24@LapDeSantiago:~$ git clone https://github.com/dsosa114/movilidad_inteligente.git Workspaces/movilidad_ws/src/
Cloning into 'Workspaces/movilidad_ws/src'...
remote: Enumerating objects: 303, done.
remote: Counting objects: 100% (303/303), done.
remote: Compressing objects: 100% (186/186), done.
remote: Total 303 (delta 118), reused 262 (delta 90), pack-reused 0 (from 0)
Receiving objects: 100% (303/303), 10.92 MiB | 5.79 MiB/s, done.
Resolving deltas: 100% (118/118), done.
sourcer_a24@LapDeSantiago:~$ cd Workspaces/movilidad_ws/src/
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws/src$ cd ..
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ls
src
```

The next is to build the packages

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ colcon build --symlink-install
Starting >>> ackermann_vehicle_bringup
Starting >>> ackermann_vehicle_description
Starting >>> control_helpers_pkg
Starting >>> differential_robot_bringup
Starting >>> differential_robot_description
Starting >>> prius_bringup
Starting >>> qcar_description
Starting >>> qcar_gazebo
Starting >>> vision_helpers_pkg
Finished <<< control_helpers_pkg [3.69s]
Finished <<< vision_helpers_pkg [3.68s]
Finished <<< ackermann_vehicle_bringup [4.90s]
Finished <<< ackermann_vehicle_description [5.00s]
Finished <<< differential_robot_bringup [5.01s]
Finished <<< differential_robot_description [5.04s]
Finished <<< qcar_description [5.06s]
Finished <<< prius_bringup [5.89s]
Finished <<< qcar_gazebo [21.1s]

Summary: 9 packages finished [21.6s]
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ls
build  install  log  src
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ source install/setup.bash
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py
```

As you can see, when we enlist one more time, you will see more elements

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ls
build  install  log  src
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ source install/setup.bash
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py
[INFO] [launch]: All log files can be found below /home/sourcer_a24/.ros/log/2025-10-11-11-57-25-944770-LapDeSantiago-20404
[INFO] [launch]: Default logging verbosity is set to INFO
[ERROR] [launch]: Caught exception in launch (see debug for traceback): Caught multiple exceptions when trying to load file of format
[py]:
 - PackageNotFoundError: "package 'ros_gz_sim' not found, searching: ['/home/sourcer_a24/Workspaces/movilidad_ws/install/vision_helper
s_pkg', '/home/sourcer_a24/Workspaces/movilidad_ws/install/qcar_gazebo', '/home/sourcer_a24/Workspaces/movilidad_ws/install/qcar_descr
iption', '/home/sourcer_a24/Workspaces/movilidad_ws/install/prius_bringup', '/home/sourcer_a24/Workspaces/movilidad_ws/install/differe
ntial_robot_description', '/home/sourcer_a24/Workspaces/movilidad_ws/install/differential_robot_bringup', '/home/sourcer_a24/Workspace
s/movilidad_ws/install/control_helpers_pkg', '/home/sourcer_a24/Workspaces/movilidad_ws/install/ackermann_vehicle_description', '/home
/sourcer_a24/Workspaces/movilidad_ws/install/ackermann_vehicle_bringup', '/opt/ros/jazzy']"
 - InvalidFrontendLaunchFileError: The launch file may have a syntax error, or its format is unknown
```

You will notice an error in the console, it's because you need the dependencies, WSL will ask for your password to access the super user

Next, we need to install the dependencies in the GitHub repository [dsosa114/movilidad_inteligente: Curso de movilidad inteligente]

Dependencies

```
sudo apt update &&
sudo apt install -y ros-${ROS_DISTRO}-ros2-control \
                    ros-${ROS_DISTRO}-ros2-controllers \
                    ros-${ROS_DISTRO}-gz-ros2-control \
                    ros-${ROS_DISTRO}-ros-gz \
                    ros-${ROS_DISTRO}-ros-gz-bridge \
                    ros-${ROS_DISTRO}-joint-state-publisher \
                    ros-${ROS_DISTRO}-robot-state-publisher \
                    ros-${ROS_DISTRO}-xacro \
                    ros-${ROS_DISTRO}-joy
```

Code:

```
 sudo apt update &&
sudo apt install -y ros-${ROS_DISTRO}-ros2-control \
ros-${ROS_DISTRO}-ros2-controllers \
ros-${ROS_DISTRO}-gz-ros2-control \
ros-${ROS_DISTRO}-ros-gz \
ros-${ROS_DISTRO}-ros-gz-bridge \
ros-${ROS_DISTRO}-joint-state-publisher \
ros-${ROS_DISTRO}-robot-state-publisher \
ros-${ROS_DISTRO}-xacro \
ros-${ROS_DISTRO}-joy
```

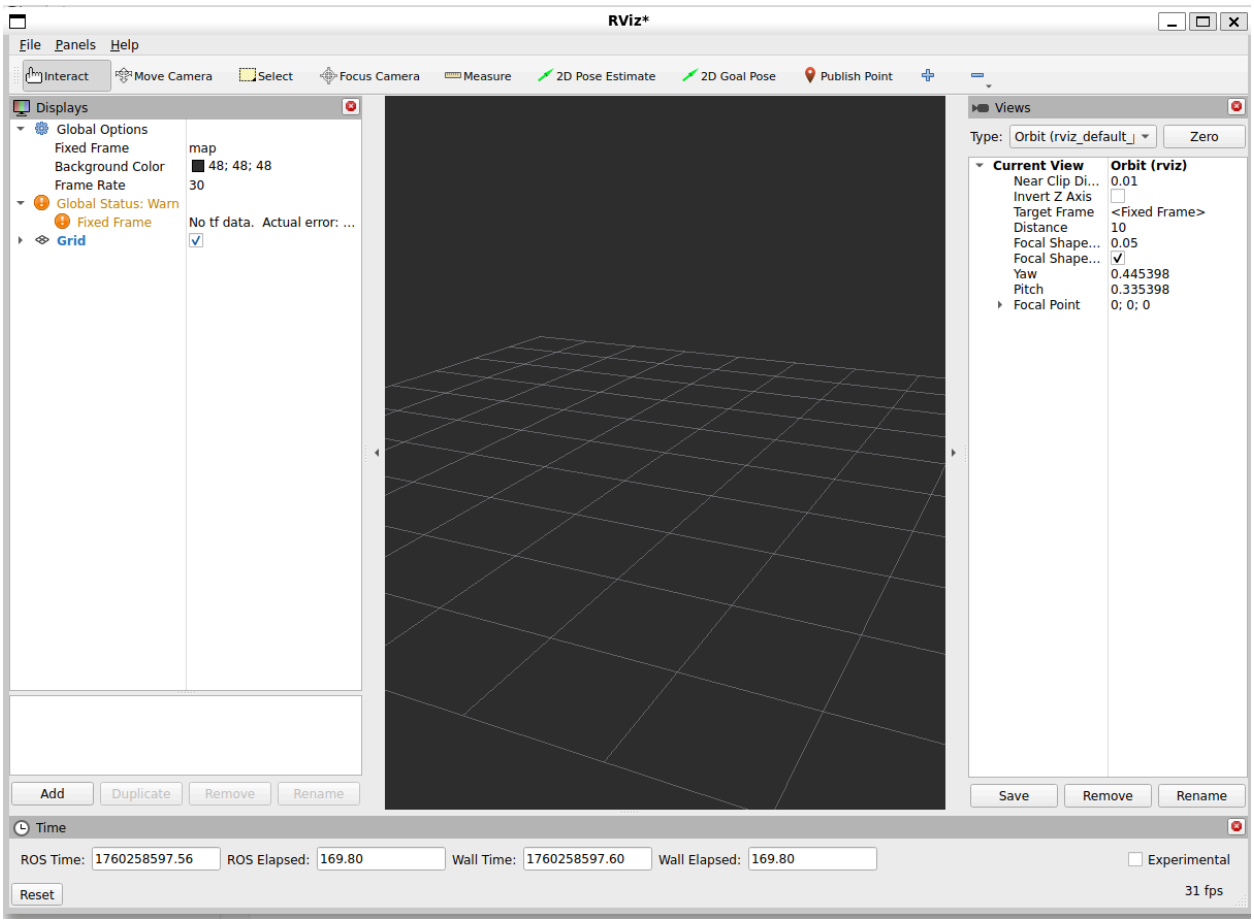
For an error that I didn't catch during the installation, my error it's different, as maybe you can see

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py
[INFO] [launch]: All log files can be found below /home/sourcer_a24/.ros/log/2025-10-11-12-06-58-897707-LapDeSantiago-23588
[INFO] [launch]: Default logging verbosity is set to INFO
[ERROR] [launch]: Caught exception in launch (see debug for traceback): 'NoneType' object has no attribute 'lower'
```

You should not have the same error, but if you have it, we can figure it out after this

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 run rviz2 rviz2
QStandardPaths: wrong permissions on runtime directory /run/user/1000/, 0755 instead of 0700
[INFO] [1760206321.445080428] [rviz2]: Stereo is NOT SUPPORTED
[INFO] [1760206321.445268604] [rviz2]: OpenGl version: 4.5 (GLSL 4.5)
[INFO] [1760206321.534178919] [rviz2]: Stereo is NOT SUPPORTED
^C[INFO] [1760206347.309705088] [rclcpp]: signal_handler(signum=2)
```

This will open ROS2 and you should be able to see the next



I tried one more time, but the error persists

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_description display.launch.py
[INFO] [launch]: All log files can be found below /home/sourcer_a24/.ros/log/2025-10-11-12-20-34-935062-LapDeSantiago-25018
[INFO] [launch]: Default logging verbosity is set to INFO
[ERROR] [launch]: Caught exception in launch (see debug for traceback): "package 'joint_state_publisher_gui' not found, searching: ['/
home/sourcer_a24/Workspaces/movilidad_ws/install/vision_helpers_pkg', '/home/sourcer_a24/Workspaces/movilidad_ws/install/qcar_gazebo',
 '/home/sourcer_a24/Workspaces/movilidad_ws/install/qcar_description', '/home/sourcer_a24/Workspaces/movilidad_ws/install/prius_bringu
p', '/home/sourcer_a24/Workspaces/movilidad_ws/install/differential_robot_description', '/home/sourcer_a24/Workspaces/movilidad_ws/ins
tall/differential_robot_bringup', '/home/sourcer_a24/Workspaces/movilidad_ws/install/control_helpers_pkg', '/home/sourcer_a24/Workspac
es/movilidad_ws/install/ackermann_vehicle_description', '/home/sourcer_a24/Workspaces/movilidad_ws/install/ackermann_vehicle_bringup',
 '/opt/ros/jazzy']"
```

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_description display.launch.py
[INFO] [launch]: All log files can be found below /home/sourcer_a24/.ros/log/2025-10-11-12-20-34-935062-LapDeSantiago-25018
[INFO] [launch]: Default logging verbosity is set to INFO
[ERROR] [launch]: Caught exception in launch (see debug for traceback): "package 'joint_state_publisher_gui' not found, searching: ['/
home/sourcer_a24/Workspaces/movilidad_ws/install/vision_helpers_pkg', '/home/sourcer_a24/Workspaces/movilidad_ws/install/qcar_gazebo',
 '/home/sourcer_a24/Workspaces/movilidad_ws/install/qcar_description', '/home/sourcer_a24/Workspaces/movilidad_ws/install/prius_bringu
p', '/home/sourcer_a24/Workspaces/movilidad_ws/install/differential_robot_description', '/home/sourcer_a24/Workspaces/movilidad_ws/ins
tall/differential_robot_bringup', '/home/sourcer_a24/Workspaces/movilidad_ws/install/control_helpers_pkg', '/home/sourcer_a24/Workspac
es/movilidad_ws/install/ackermann_vehicle_description', '/home/sourcer_a24/Workspaces/movilidad_ws/install/ackermann_vehicle_bringup',
 '/opt/ros/jazzy']"
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ sudo apt install ros-jazzy-joint-state-publisher-gui
[sudo] password for sourcer_a24:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  ros-jazzy-joint-state-publisher-gui
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 16.8 kB of archives.
After this operation, 73.7 kB of additional disk space will be used.
Get:1 http://packages.ros.org/ros2/ubuntu noble/main amd64 ros-jazzy-joint-state-publisher-gui amd64 2.4.0-3noble.20250824.090333 [16.
8 kB]
Fetched 16.8 kB in 7s (2,444 B/s)
Selecting previously unselected package ros-jazzy-joint-state-publisher-gui.
(Reading database ... 159250 files and directories currently installed.)
Preparing to unpack .../ros-jazzy-joint-state-publisher-gui_2.4.0-3noble.20250824.090333_amd64.deb ...
Unpacking ros-jazzy-joint-state-publisher-gui (2.4.0-3noble.20250824.090333) ...
Setting up ros-jazzy-joint-state-publisher-gui (2.4.0-3noble.20250824.090333) ...
```

I tried to solve it by installing the libraries

But the problem lies in the last try on the next capture

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_description display.launch.py
[INFO] [launch]: All log files can be found below /home/sourcer_a24/.ros/log/2025-10-11-12-21-55-463615-LapDeSantiago-25347
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [25351]
[INFO] [joint_state_publisher_gui-2]: process started with pid [25352]
[INFO] [rviz2-3]: process started with pid [25353]
[robot_state_publisher-1] terminate called after throwing an instance of 'pluginlib::LibraryLoadException'
[robot_state_publisher-1]   what():  Failed to load library /opt/ros/jazzy/lib/libsdformat_urdf_plugin.so. Make sure that you are call
ing the PLUGINLIB_EXPORT_CLASS macro in the library code, and that names are consistent between this macro and your XML. Error string:
 Could not load library dlopen error: libsdformat14.so.14: cannot open shared object file: No such file or directory, at ./src/shared_
library.c:99
[rviz2-3] QStandardPaths: wrong permissions on runtime directory /run/user/1000/, 0755 instead of 0700
[ERROR] [robot_state_publisher-1]: process has died [pid 25351, exit code -6, cmd '/opt/ros/jazzy/lib/robot_state_publisher/robot_stat
e_publisher --ros-args -r __node:=robot_state_publisher --params-file /tmp/launch_params_4i8az8h4'].
[joint_state_publisher_gui-2] QStandardPaths: wrong permissions on runtime directory /run/user/1000/, 0755 instead of 0700
[joint_state_publisher_gui-2] [INFO] [1760206916.044638197] [joint_state_publisher_gui]: Waiting for robot_description to be published
 on the robot_description topic...
[joint_state_publisher_gui-2] [INFO] [1760206916.154782526] [joint_state_publisher_gui]: Centering
[rviz2-3] [INFO] [1760206916.798197397] [rviz2]: Stereo is NOT SUPPORTED
[rviz2-3] [INFO] [1760206916.798587506] [rviz2]: OpenGl version: 4.5 (GLSL 4.5)
[rviz2-3] [INFO] [1760206917.001127838] [rviz2]: Stereo is NOT SUPPORTED
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
[ERROR] [joint_state_publisher_gui-2]: process has died [pid 25352, exit code -2, cmd '/opt/ros/jazzy/lib/joint_state_publisher_gui/jo
int_state_publisher_gui --ros-args -r __node:=joint_state_publisher_gui'].
[INFO] [rviz2-3]: process has finished cleanly [pid 25353]
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ gz
gz: command not found
```

When you press gz you should see something like

```
sourcer_a24@LapDeSantiago:~$ gz
The 'gz' command provides a command line interface to the Gazebo Tools.

  gz <command> [options]

List of available commands:

  help:          Print this help text.
  fuel:          Manage simulation resources.
  gui:           Launch graphical interfaces.
  log:           Record or playback topics.
  model:         Print information about models.
  msg:           Print information about messages.
  param:         List, get or set parameters.
  plugin:        Print information about plugins.
  sdf:           Utilities for SDF files.
  service:       Print information about services.
  sim:           Run and manage the Gazebo Simulator.
  topic:         Print information about topics.

Options:

  --force-version <VERSION>  Use a specific library version.
  --versions                 Show the available versions.
  --commands                 Show the available commands.
Use 'gz help <command>' to print help for a command.
sourcer_a24@LapDeSantiago:~$
```

To solve this I goes to the next link:
https://gazebosim.org/docs/harmonic/install_ubuntu/

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ gz
gz: command not found
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ sudo apt-get update
sudo apt-get install curl lsb-release gnupg
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://packages.ros.org/ros2/ubuntu noble InRelease [4,676 B]
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 http://packages.ros.org/ros2/ubuntu noble/main Sources [1,810 kB]
Get:7 http://packages.ros.org/ros2/ubuntu noble/main amd64 Packages [1,739 kB]
Fetched 3,554 kB in 8s (437 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
lsb-release is already the newest version (12.0-2).
lsb-release set to manually installed.
gnupg is already the newest version (2.4.4-2ubuntu17.3).
gnupg set to manually installed.
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ sudo apt-get update
sudo apt-get install curl lsb-release gnupg^C
```

Note for myself: The ^C was because I did it already in the last prompt

Then the installation starts

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ sudo apt-get update
sudo apt-get install curl lsb-release gnupg^C
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ sudo curl https://packages.osrfoundation.org/gazebo.gpg --output /usr/share/keyri
ngs/pkgs-osrf-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/pkgs-osrf-archive-keyring.gpg] https://packages.osrfoundati
on.org/gazebo/ubuntu-stable $(lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/gazebo-stable.list > /dev/null
sudo apt-get update
sudo apt-get install gz-harmonic
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  1222  100  1222    0     0   2062      0 --:--:-- --:--:-- --:--:--  2064
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://packages.ros.org/ros2/ubuntu noble InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 https://packages.osrfoundation.org/gazebo/ubuntu-stable noble InRelease [4,267 B]
Get:7 https://packages.osrfoundation.org/gazebo/ubuntu-stable noble/main amd64 Packages [95.3 kB]
Fetched 99.6 kB in 7s (13.5 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  gz-launch7-cli gz-plugin2-cli gz-sim8-cli gz-tools2 gz-transport13-cli libdart-core+collisions+odelcpsolver6.13 libev-dev libev4t64
libgz-cmake3-dev libgz-common5
  libgz-common5-av libgz-common5-av-dev libgz-common5-core-dev libgz-common5-dev libgz-common5-events libgz-common5-events-dev libgz-c
ommon5-geospatial
  libgz-common5-geospatial-dev libgz-common5-graphics libgz-common5-graphics-dev libgz-common5-io libgz-common5-io-dev libgz-common5-p
rofiler libgz-common5-profiler-dev
```

For a reason that I not understand this happened

```
[parameter_bridge-4] [INFO] [1760207493.177588681] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/world/empty/model/differential_robot/joint_state (gz.msgs.Model) -> /joint_states (sensor_msgs/msg/JointState)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760207493.196017353] [ros_gz_bridge]: Creating ROS->GZ Bridge: [/cmd_vel (geometry_msgs/msg/Twist) -> /model/differential_robot/cmd_vel (gz.msgs.Twist)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760207493.206318952] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/scan (gz.msgs.LaserScan) -> /scan (sensor_msgs/msg/LaserScan)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760207493.215218871] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/camera/image_raw (gz.msgs.Image) -> /camera/image_raw (sensor_msgs/msg/Image)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760207493.224705250] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/camera/camera_info (gz.msgs.CameraInfo) -> /camera/camera_info (sensor_msgs/msg/CameraInfo)] (Lazy 0)
[create-3] [INFO] [1760207497.080000555] [ros_gz_sim]: Requesting list of world names.
[create-3] [INFO] [1760207502.082789824] [ros_gz_sim]: Requesting list of world names.
[create-3] [INFO] [1760207507.084451152] [ros_gz_sim]: Requesting list of world names.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining table/1/meshes/diningtable.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining table/1/meshes/diningtable.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining chair/1/meshes/diningchair.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining chair/1/meshes/diningchair.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining chair/1/meshes/diningchair.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/refrigerator/1/meshes/fridge.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 9: Msg: Failed to load a world.
[create-3] [INFO] [1760207509.606547208] [ros_gz_sim]: Requesting list of world names.
[INFO] [gazebo-2]: process has finished cleanly [pid 28932]
[create-3] [INFO] [1760207514.609873905] [ros_gz_sim]: Requesting list of world names.
[create-3] [INFO] [1760207519.612300080] [ros_gz_sim]: Requesting list of world names.
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
```

```
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining table/1/meshes/diningtable.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining table/1/meshes/diningtable.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining chair/1/meshes/diningchair.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining chair/1/meshes/diningchair.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/dining chair/1/meshes/diningchair.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 14: Msg: Parser configurations requested resolved uris, but uri [file:///home/daniel/.ignition/fuel/fuel.gazebosim.org/openrobotics/models/refrigerator/1/meshes/fridge.dae] could not be resolved.
[gazebo-2] [Err] [Server.cc:86] Error Code 9: Msg: Failed to load a world.
[create-3] [INFO] [1760207509.606547208] [ros_gz_sim]: Requesting list of world names.
[INFO] [gazebo-2]: process has finished cleanly [pid 28932]
[create-3] [INFO] [1760207514.609873905] [ros_gz_sim]: Requesting list of world names.
[create-3] [INFO] [1760207519.612300080] [ros_gz_sim]: Requesting list of world names.
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
[rviz2-5] [INFO] [1760207523.223557570] [rclcpp]: signal_handler(signum=2)
[create-3] [INFO] [1760207523.223708971] [rclcpp]: signal_handler(signum=2)
[parameter_bridge-4] [INFO] [1760207523.223679105] [rclcpp]: signal_handler(signum=2)
[robot_state_publisher-1] [INFO] [1760207523.223914882] [rclcpp]: signal_handler(signum=2)
[INFO] [parameter_bridge-4]: process has finished cleanly [pid 28935]
[INFO] [robot_state_publisher-1]: process has finished cleanly [pid 28931]
[INFO] [rviz2-5]: process has finished cleanly [pid 28936]
[create-3] [INFO] [1760207524.616683469] [ros_gz_sim]: Timed out when getting world names.
[ERROR] [create-3]: process has died [pid 28933, exit code 255, cmd '/opt/ros/jazzy/lib/ros_gz_sim/create -topic /robot_description --ros-args'].
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py gz_args:=
build/   install/ log/     src/
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py gz_args:=´empty.sdf -r´
```

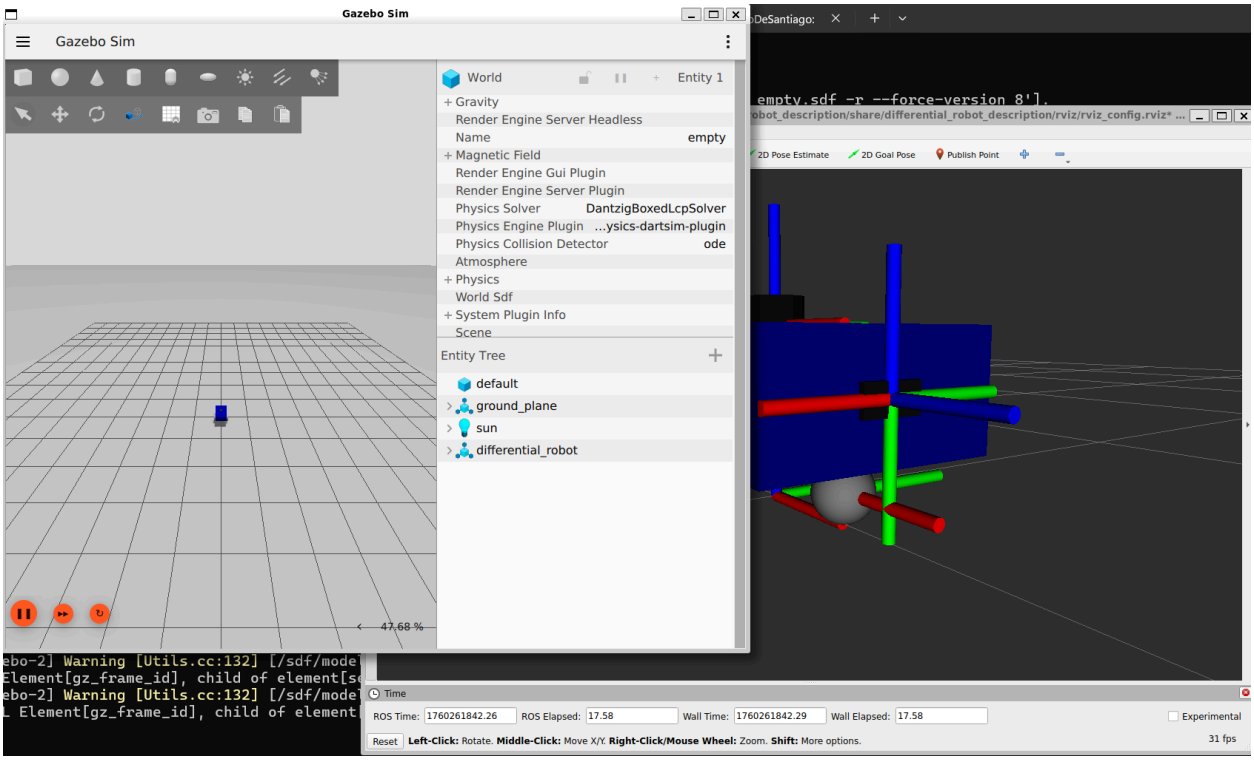
```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py gz_args:=
build/   install/ log/     src/
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py gz_args:=´empty.sdf -r´
usage: ros2 [-h] [--use-python-default-buffering] Call `ros2 <command> -h` for more detailed usage. ...
ros2: error: unrecognized arguments: -r´
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py gz_args:="empty.sdf -r"
[INFO] [launch]: All log files can be found below /home/sourcer_a24/.ros/log/2025-10-11-12-35-32-484192-LapDeSantiago-29765
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [29770]
[INFO] [gazebo-2]: process started with pid [29771]
[INFO] [create-3]: process started with pid [29772]
[INFO] [parameter_bridge-4]: process started with pid [29774]
[INFO] [rviz2-5]: process started with pid [29775]
[robot_state_publisher-1] [INFO] [1760207733.245122002] [robot_state_publisher]: Robot initialized
[rviz2-5] QStandardPaths: wrong permissions on runtime directory /run/user/1000/, 0755 instead of 0700
[create-3] [INFO] [1760207733.257609163] [ros_gz_sim]: Requesting list of world names.
[rviz2-5] [INFO] [1760207733.492908789] [rviz2]: Stereo is NOT SUPPORTED
[rviz2-5] [INFO] [1760207733.493271634] [rviz2]: OpenGl version: 4.5 (GLSL 4.5)
[gazebo-2] QStandardPaths: wrong permissions on runtime directory /run/user/1000/, 0755 instead of 0700
[rviz2-5] [INFO] [1760207733.555863076] [rviz2]: Stereo is NOT SUPPORTED
[create-3] [INFO] [1760207734.190128990] [ros_gz_sim]: Waiting messages on topic [/robot_description].
[create-3] [INFO] [1760207734.290859361] [ros_gz_sim]: Entity creation successful.
[parameter_bridge-4] [INFO] [1760207734.296287585] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/clock (gz.msgs.Clock) -> /clock (rosgra
ph_msgs/msg/Clock)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760207734.305036714] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/model/differential_robot/tf (gz.msgs.Po
se_V) -> /tf (tf2_msgs/msg/TFMessage)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760207734.317595802] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/world/empty/model/differential_robot/jo
int_state (gz.msgs.Model) -> /joint_states (sensor_msgs/msg/JointState)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760207734.322358254] [ros_gz_bridge]: Creating ROS->GZ Bridge: [/cmd_vel (geometry_msgs/msg/Twist) -> /m
odel/differential_robot/cmd_vel (gz.msgs.Twist)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760207734.328429376] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/scan (gz.msgs.LaserScan) -> /scan (sens
or_msgs/msg/LaserScan)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760207734.333860456] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/camera/image_raw (gz.msgs.Image) -> /ca
```

```
[robot_state_publisher-1] [WARN] [1760208611.044235639] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760208640.813696276] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760208670.579252824] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760208700.344542127] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760208730.115305263] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760208759.879729008] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760208789.658673995] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760208819.423893329] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760208849.187432631] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760209200.345920713] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760209230.991705482] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
[robot_state_publisher-1] [WARN] [1760209260.693710915] [robot_state_publisher]: Moved backwards in time, re-publishing joint transfor
ms!
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)
[rviz2-5] [INFO] [1760214228.680936109] [rclcpp]: signal_handler(signum=2)
[parameter_bridge-4] [INFO] [1760214228.681250583] [rclcpp]: signal_handler(signum=2)
[robot_state_publisher-1] [INFO] [1760214228.681252255] [rclcpp]: signal_handler(signum=2)
[INFO] [robot_state_publisher-1]: process has finished cleanly [pid 29770]
[INFO] [parameter_bridge-4]: process has finished cleanly [pid 29774]
[INFO] [rviz2-5]: process has finished cleanly [pid 29775]
[ERROR] [gazebo-2]: process has died [pid 29771, exit code -2, cmd 'ruby /usr/bin/gz sim empty.sdf -r --force-version 8'].
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py gz_args:="empty.sdf -r"
```

Tests:

```
sourcer_a24@LapDeSantiago:~/Workspaces/movilidad_ws$ ros2 launch differential_robot_bringup gz_sim.launch.py gz_args:="empty.sdf -r"
[INFO] [launch]: All log files can be found below /home/sourcer_a24/.ros/log/2025-10-12-03-27-01-427879-LapDeSantiago-70342
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [70346]
[INFO] [gazebo-2]: process started with pid [70347]
[INFO] [create-3]: process started with pid [70348]
[INFO] [parameter_bridge-4]: process started with pid [70350]
[INFO] [rviz2-5]: process started with pid [70351]
[rviz2-5] QStandardPaths: wrong permissions on runtime directory /run/user/1000/, 0755 instead of 0700
[robot_state_publisher-1] [INFO] [1760261222.219284073] [robot_state_publisher]: Robot initialized
[create-3] [INFO] [1760261222.273690164] [ros_gz_sim]: Requesting list of world names.
[rviz2-5] [INFO] [1760261222.433312520] [rviz2]: Stereo is NOT SUPPORTED
[rviz2-5] [INFO] [1760261222.433602983] [rviz2]: OpenGl version: 4.5 (GLSL 4.5)
[rviz2-5] [INFO] [1760261222.487407535] [rviz2]: Stereo is NOT SUPPORTED
[gazebo-2] QStandardPaths: wrong permissions on runtime directory /run/user/1000/, 0755 instead of 0700
[create-3] [INFO] [1760261223.303850849] [ros_gz_sim]: Waiting messages on topic [/robot_description].
[parameter_bridge-4] [INFO] [1760261223.323187165] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/clock (gz.msgs.Clock) -> /clock (rosgraph_msgs/msg/Clock)] (L
azy 0)
[parameter_bridge-4] [INFO] [1760261223.330778387] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/model/differential_robot/tf (gz.msgs.Pose_V) -> /tf (tf2_msgs
/msg/TFMessage)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760261223.348193871] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/world/empty/model/differential_robot/joint_state (gz.msgs.Mod
el) -> /joint_states (sensor_msgs/msg/JointState)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760261223.355927596] [ros_gz_bridge]: Creating ROS->GZ Bridge: [/cmd_vel (geometry_msgs/msg/Twist) -> /model/differential_robo
t/cmd_vel (gz.msgs.Twist)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760261223.366129696] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/scan (gz.msgs.LaserScan) -> /scan (sensor_msgs/msg/LaserScan)
] (Lazy 0)
[parameter_bridge-4] [INFO] [1760261223.367894524] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/camera/image_raw (gz.msgs.Image) -> /camera/image_raw (sensor
_msgs/msg/Image)] (Lazy 0)
[parameter_bridge-4] [INFO] [1760261223.369542420] [ros_gz_bridge]: Creating GZ->ROS Bridge: [/camera/camera_info (gz.msgs.CameraInfo) -> /camera/camera_inf
o (sensor_msgs/msg/CameraInfo)] (Lazy 0)
[create-3] [INFO] [1760261223.402751081] [ros_gz_sim]: Entity creation successful.
[INFO] [create-3]: process has finished cleanly [pid 70348]
[gazebo-2] Warning [Utils.cc:132] [/sdf/model[@name="differential_robot"]/link[@name="base_footprint"]/sensor[@name="lidar"]/gz_frame_id:<urdf-string>:L0]:
XML Element[gz_frame_id], child of element[sensor], not defined in SDF. Copying[gz_frame_id] as children of [sensor].
[gazebo-2] Warning [Utils.cc:132] [/sdf/model[@name="differential_robot"]/link[@name="base_footprint"]/sensor[@name="lidar"]/gz_frame_id:<data-string>:L189]
: XML Element[gz_frame_id], child of element[sensor], not defined in SDF. Copying[gz_frame_id] as children of [sensor].
```

Your computer should open the next:



To the next prof you need to start a new tab into your cmd, if you close the earlier, you don’t gonna be able to test the next, if you already did it, don’t problem just open it again and put the same last instruction

```
Microsoft Windows [Versión 10.0.26200.6725]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\ismae>wsl
sourcer_a24@LapDeSantiago:/mnt/c/Users/ismae$ cd
sourcer_a24@LapDeSantiago:~$ ros2 run teleop_twist_keyboard  teleop_twist_keyboard

This node takes keypresses from the keyboard and publishes them
as Twist/TwistStamped messages. It works best with a US keyboard layout.
---------------------------
Moving around:
   u    i    o
   j    k    l
   m    ,    .

For Holonomic mode (strafing), hold down the shift key:
---------------------------
   U    I    O
   J    K    L
   M    <    >

t : up (+z)
b : down (-z)

anything else : stop

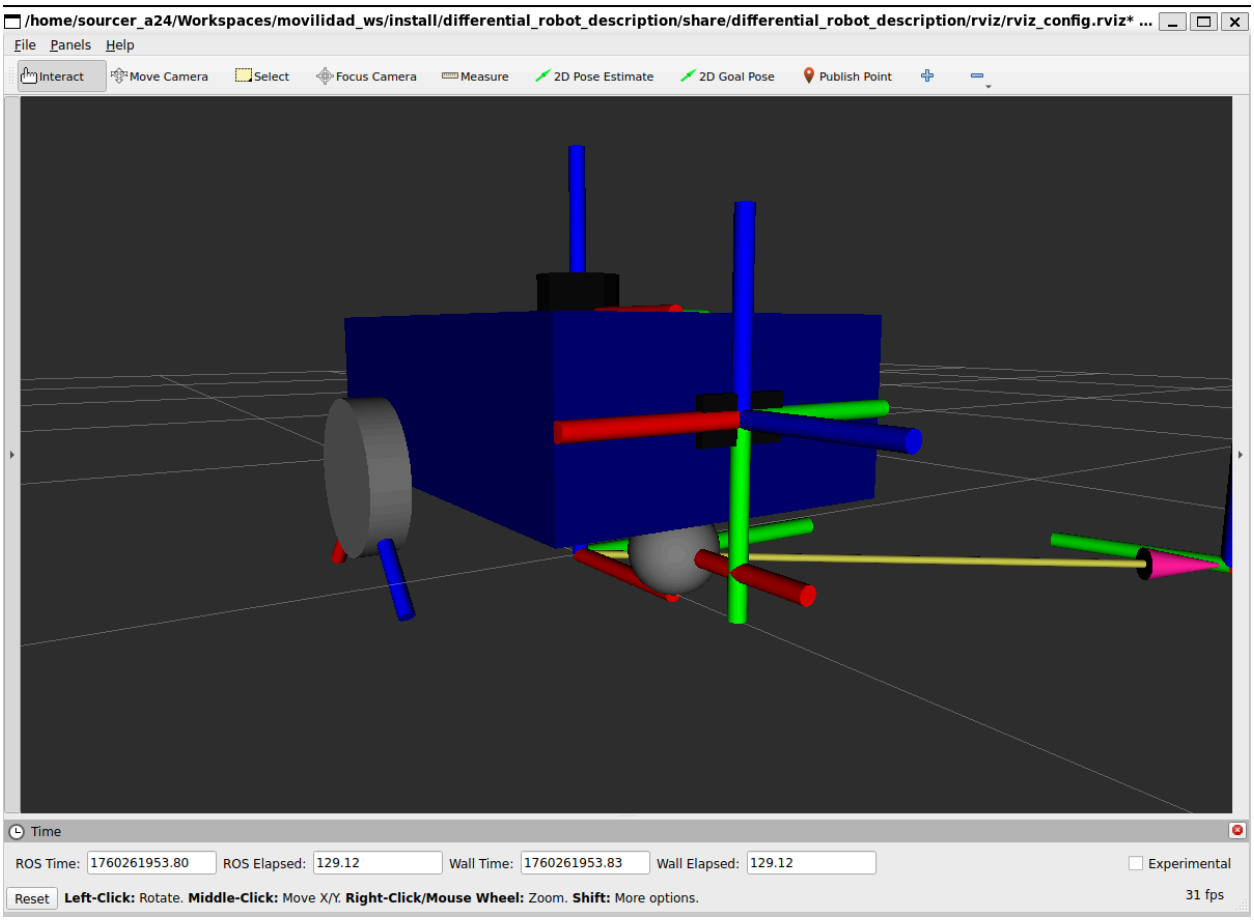
q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

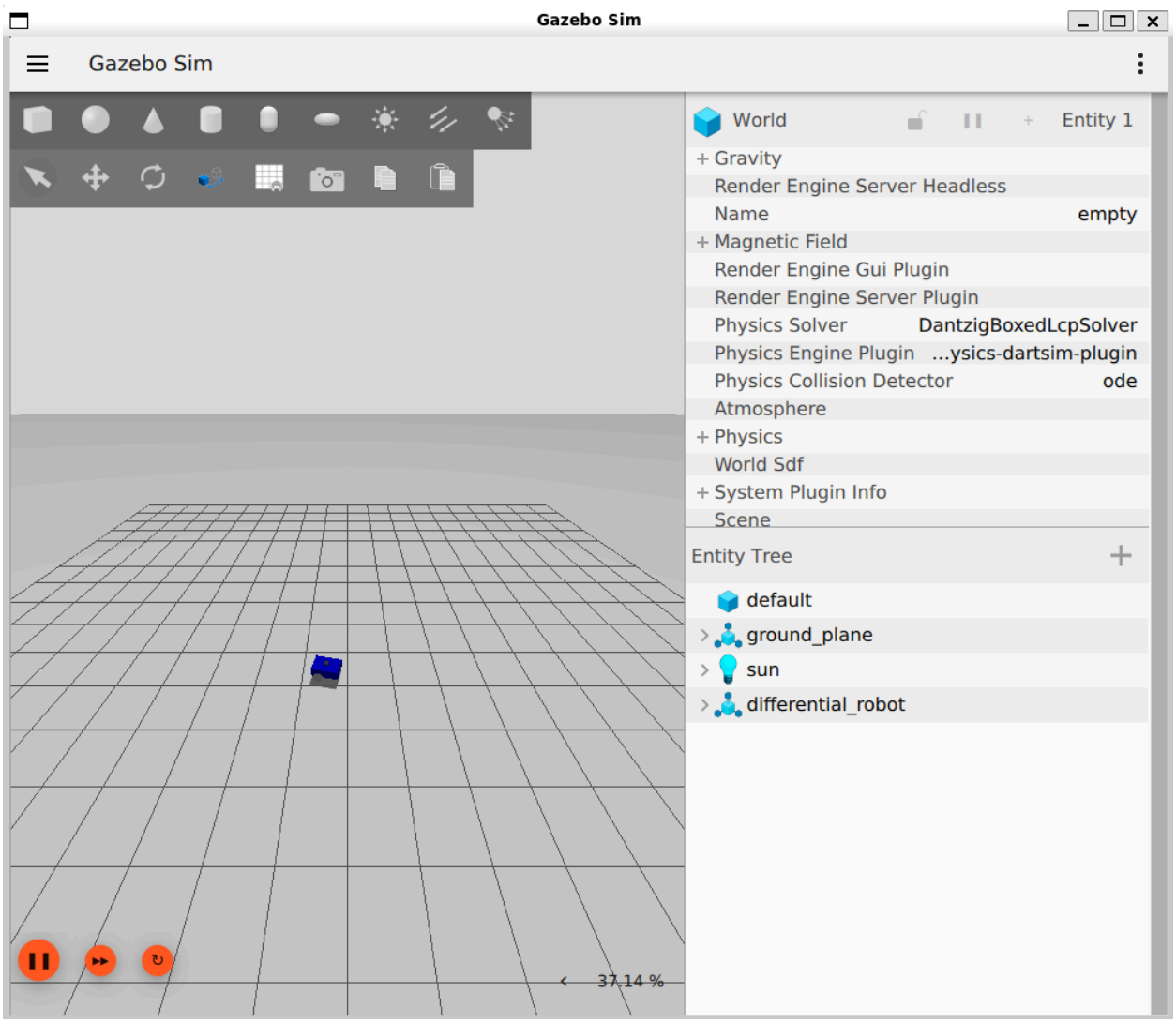
CTRL-C to quit

sourcer_a24@LapDeSantiago:~$ ros2 run teleop_twist_keyboard  teleop_twist_keyboard
```

If you press u for example, you gonna look your robot/model looking around in both tabs

This should deploy the nextBut after all with the last prompt you should see this


/home/sourcer_a24/Workspaces/movilidad_ws/install/differential_robot_description/share/differential_robot_description/rviz/rviz_config.rviz* ...
File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
Time
ROS Time: 1760261953.80
ROS Elapsed: 129.12
Wall Time: 1760261953.83
Wall Elapsed: 129.12
Experimental
Reset
Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click/Mouse Wheel: Zoom. Shift: More options.
31 fps


Gazebo Sim
Gazebo Sim
World
Entity 1
+ Gravity
Render Engine Server Headless
Name
empty
+ Magnetic Field
Render Engine Gui Plugin
Render Engine Server Plugin
Physics Solver
DantzigBoxedLcpSolver
Physics Engine Plugin
...ysics-dartsim-plugin
Physics Collision Detector
ode
Atmosphere
+ Physics
World Sdf
+ System Plugin Info
Scene
Entity Tree
default
ground_plane
sun
differential_robot
37.14 %